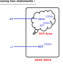
***** What is the difference between the following two statements :**

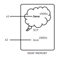
a) String s1 = "Java";
b) String s2 = new String("NIT");



* Whenever we create a String object using new keyword then most of the time two objects are created as shown in the above diagram.

* If the String object which we are creating using new keyword is already available in SCP area then only one object will be created as shown in the diagram below.

```
String s1 = "Java";
String s2 = new String("Java");
```



So, the final conclusion is :

String s1 = "Java";   //Only one Object and one reference variable will be created

String s2 = new String("Hyderabad"); //Two Objects one is in SCP & another one is in NON-SCP will be created

String s3 = new String("Java"); //Only one object i.e. Java will be created in a non scp area because SCP cannot have duplicate objects.

How JVM is going to place the String object in the SCP area which is created by using String literal :

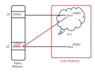
* In order to place the String Object which is created by using String Literal ("") JVM internally calling **intern()** method.

public String intern() :

* It is a predefined non static method of String class.
* It is used to return the canonical representation of the String.

```
void main()
{
    String s1 = "Java";
    String s2 = new String("Java");

    IO.println(s1==s2); //false
    s2 = s2.intern();
    IO.println(s1==s2); //true
}
```

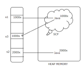



```
String s1 = "Java";
String s2 = new String("Java");

IO.println(s1==s2);  //false

String s3 = s2.intern();

IO.println(s1==s2);  //false
```



When we call intern() on a String object created by using new keyword, It will place the object in the SCP area (if not available) and returns the reference from SCP.

If the String is already existing in the SCP area then It will return the reference of the existing String object from the SCP.

Note :- JVM internally uses intern process to place the String literal ("") inside SCP area [interning process]

How many ways we can put the String object into SCP area :

The following are the ways to put the String object into SCP area

1) By using Literal
2) By using intern() method of new Object
3) By using valueOf() static method of String class.
4) By using '+' operator [Compile time constant]

```
void main()
{
    String s1 = "Java technology";
    String s2 = new String("Java technology");
    s2 = s2.intern();

    IO.println(s1==s2);

    String s3 = String.valueOf("Java technology");
    IO.println(s1==s3);

    String s4 = "Java" + " technology";
    IO.println(s1==s4);
}
```

Working with methods of String :-

String class has provided number of predefined methods to work with String which are as follows :-

1) public char charAt(int indexPosition) :-

It is used to extract a character based on the index position. If index is not available It will return StringIndexOutOfBoundsException

```
void main()
{
    String s1 = "Java";
    char ch = s1.charAt(0);
    IO.println(ch);
}
```

public String concat(String str) :-

It is used to concatenate two Strings. This can be also done by using concatenation operator '+'.

```
void main()
{
    String s1 = "Java";
    s1 = s1.concat(" technology");
    IO.println(s1);

    String s2 = "Data";
    String s3 = "base";
    s3 = s2 + s3;
    IO.println(s3);


    String s4 = "Tata";
    String s5 = "Nagar";
    String s6 = s4.concat(s5);
    IO.println(s6);
}
```

public boolean equals(Object obj)

It is used to verify whether two String Objects are equal or not based on the **content.**

It takes Object as a parameter because it is an **overridden** method.It is overridden from Object class.

```
void main()
{
    String s1 = "Java";
    String s2 = new String("Java");

    System.out.println(s1==s2);
    System.out.println(s1.equals(s2));

}
```